

# Automated Usability Evaluation in Virtual Reality: A Systematic Mapping Study - Supplementary Material

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We used specific search queries for the digital libraries, but with common logic:

```
Title OR Keywords OR Abstract
should match
Virtual Reality AND usability AND evaluation
```

Including combinations of OR connected synonyms for *Virtual Reality*, *usability*, and *evaluation*.

## 1 Springer Link

```
(
  "Virtual Reality" OR "VR"
  OR "Augmented Reality" OR "AR"
  OR "Mixed Reality" OR "XR"
  OR "Extended Reality"
)
AND (
  "usability" OR "user experience" OR "UX"
)
AND (
  "evaluating" OR "testing" OR "evaluation"
  OR "test" OR "validation" OR "validating"
)
```

## 2 Scopus

```
TITLE-ABS-KEY(
  (
    "Virtual Reality" OR "VR"
    OR "Augmented Reality" OR "AR"
    OR "Mixed Reality" OR "XR"
    OR "Extended Reality"
  )
  AND (
    "usability" OR "user experience" OR "UX"
  )
  AND (
    "automated" OR "automation" OR "automatic"
    OR "agent based" OR "agent-based"
  )
  AND (
    "evaluating" OR "testing" OR "evaluation"
    OR "test" OR "validation" OR "validating"
  )
)
```

```

)
AND(
    LIMIT-TO ( DOCTYPE , "cp" )
    OR LIMIT-TO ( DOCTYPE , "ar" )
)
)

3 IEEE Xplore
(
    (
        "Document Title":"Virtual Reality"
        OR "Document Title":"VR"
        OR "Document Title":"Augmented Reality"
        OR "Document Title":"AR"
        OR "Document Title":"Mixed Reality"
        OR "Document Title":"XR"
        OR "Document Title":"Extended Reality"
    )
    AND (
        "Document Title":"usability"
        OR "Document Title":"user experience"
        OR "Document Title":"UX"
    )
    AND (
        "Document Title":"evaluating"
        OR "Document Title":"testing"
        OR "Document Title":"evaluation"
        OR "Document Title":"test"
        OR "Document Title":"validation"
        OR "Document Title":"validating"
    )
)
OR (
    (
        "Author Keywords":"Virtual Reality"
        OR "Author Keywords":"VR"
        OR "Author Keywords":"Augmented Reality"
        OR "Author Keywords":"AR"
        OR "Author Keywords":"Mixed Reality"
        OR "Author Keywords":"XR"
        OR "Author Keywords":"Extended Reality"
    )
    AND (
        "Author Keywords":"usability"
        OR "Author Keywords":"user experience"
    )
)

```

```

    OR "Author Keywords": "UX"
  )
  AND (
    "Author Keywords": "evaluating"
    OR "Author Keywords": "testing"
    OR "Author Keywords": "evaluation"
    OR "Author Keywords": "test"
    OR "Author Keywords": "validation"
    OR "Author Keywords": "validating"
  )
)
OR (
  (
    "Abstract": "Virtual Reality"
    OR "Abstract": "VR"
    OR "Abstract": "Augmented Reality"
    OR "Abstract": "AR"
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    OR "Abstract": "XR"
    OR "Abstract": "Extended Reality"
  )
  AND (
    "Abstract": "usability"
    OR "Abstract": "user experience"
    OR "Abstract": "UX"
  )
  AND (
    "Abstract": "automated"
    OR "Abstract": "automation"
    OR "Abstract": "automatic"
    OR "Abstract": "agent based"
    OR "Abstract": "agent-based"
  )
  AND (
    "Abstract": "evaluating"
    OR "Abstract": "testing"
    OR "Abstract": "evaluation"
    OR "Abstract": "test"
    OR "Abstract": "validation"
    OR "Abstract": "validating"
  )
)
)

```

#### 4 The ACM Guide to Computing Literature

Title:(

```

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  AND (
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    OR "test" OR "validation" OR "validating"
  )
)
)
OR Abstract:(
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    OR "Extended Reality"
  )
  AND (
    "usability" OR "user experience" OR "UX"
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  AND (
    "automated" OR "automation" OR "automatic"
    OR "agent based" OR "agent-based")
  AND (
    "evaluating" OR "testing" OR "evaluation"
    OR "test" OR "validation" OR "validating")
)
)

```

**Table 1: Overview of all included publications and mapping to method, measurement, and evaluated usability dimension**

| Ref  | Methods                                 | Usability Dimensions                     | Measurements                                                   |
|------|-----------------------------------------|------------------------------------------|----------------------------------------------------------------|
| [1]  | automated measurement                   | physical effort                          | pressure load                                                  |
| [3]  | automated measurement                   | flow                                     | heart rate, EMG, respiration                                   |
| [4]  | automated measurement                   | usability, physical effort               | hand movement, head movement                                   |
| [8]  | automated measurement                   | attention, cognitive load                | EEG                                                            |
| [10] | automated measurement                   | physical effort                          | heart rate                                                     |
| [11] | automated measurement                   | UX                                       | heart rate, events                                             |
| [13] | automated measurement, visualization    | UX                                       | movement                                                       |
| [15] | automated measurement                   | presence                                 | heart rate                                                     |
| [14] | automated measurement                   | presence, area of interest, realism      | heart rate, EDA, gaze                                          |
| [16] | automated measurement                   | usability, behaviour, engagement         | hand movement, head movement, events, gaze                     |
| [20] | automated measurement, AutoQUEST        | usability smells                         | events                                                         |
| [21] | automated measurement                   | engagement                               | head movement, EEG                                             |
| [26] | automated measurement                   | usability                                | EEG                                                            |
| [27] | automated measurement                   | learnability                             | facial expression                                              |
| [28] | automated measurement                   | emotions                                 | head movement                                                  |
| [30] | automated measurement, machine learning | anxiety                                  | heart rate                                                     |
| [31] | automated measurement, machine learning | usability                                | hand movement, head movement, EEG, speech                      |
| [32] | automated measurement, AutoQUEST        | usability smells                         | events                                                         |
| [34] | automated measurement                   | physiological arousal                    | heart rate, EDA, body temperature                              |
| [37] | automated measurement                   | presence, flow, emotions, cognitive load | EEG, EDA                                                       |
| [38] | automated measurement                   | stress, ease of use, attention           | heart rate, gaze                                               |
| [40] | automated measurement, machine learning | attention                                | EEG, pupil                                                     |
| [41] | automated measurement                   | UX, learnability                         | decision time                                                  |
| [43] | automated measurement, machine learning | presence                                 | hand movement, head movement, speech, body tracking            |
| [42] | automated measurement, machine learning | presence                                 | hand movement, head movement, speech, body tracking            |
| [69] | automated measurement, visualization    | area of interest                         | gaze                                                           |
| [45] | automated measurement, machine learning | presence                                 | hand movement, head movement, facial expression, scene capture |
| [46] | automated measurement                   | attention, UI                            | head movement, gaze                                            |
| [47] | automated measurement                   | area of interest                         | gaze, body tracking                                            |
| [49] | agent                                   | UX, learnability, emotions               | none                                                           |
| [50] | automated generation, machine learning  | realism                                  | none                                                           |
| [51] | automated measurement, visualization    | presence, attention, UI, comfort         | gaze                                                           |
| [2]  | agent                                   | usability, emotions                      | none                                                           |
| [53] | automated measurement                   | physiological arousal                    | heart rate, EDA                                                |
| [54] | automated measurement                   | interactability, learnability            | head movement                                                  |
| [56] | automated measurement                   | stress                                   | heart rate                                                     |
| [57] | machine learning                        | behaviour, sentiment                     | events, speech                                                 |
| [58] | automated measurement                   | UX, emotions, stress                     | heart rate, facial expression, events, speech                  |
| [60] | automated measurement                   | attention, behaviour                     | head movement, EDA                                             |
| [61] | automated measurement, visualization    | flow, stress                             | head movement, pupil, movement                                 |
| [63] | automated measurement                   | stress, anxiety                          | heart rate, walking speed                                      |
| [65] | automated measurement                   | usability, physical effort               | hand movement, head movement, EMG, pressure load               |